

Jinlin Lai

ASSISTANT PROFESSOR · PROBABILISTIC MACHINE LEARNING

✉ jilai@csusm.edu | 🏠 lll6924.github.io

Positions

California State University San Marcos

ASSISTANT PROFESSOR IN COMPUTER SCIENCE

San Marcos, CA

August 2026 - Now

Basis AI Institute

SUMMER RESEARCH INTERN

New York, NY

June 2025 - August 2025

Flatiron Institute, Simons Foundation

SUMMER PRE-DOCTORAL RESEARCHER

New York, NY

May 2024 - August 2024

Dolby Laboratories Inc.

ATG IMAGING RESEARCH INTERN

Sunnyvale, CA

June 2023 - August 2023

Education

University of Massachusetts Amherst

PH.D. IN COMPUTER SCIENCE

Amherst, MA, United States

Aug 2020 - June 2026

- Advisor: Dr. Daniel R. Sheldon
- Area: Probabilistic machine learning, computational statistics and Bayesian inference

University of Massachusetts Amherst

M.S. IN COMPUTER SCIENCE

Amherst, MA, United States

Aug 2020 - Feb 2024

Tsinghua University

B.ENG. OF COMPUTER SCIENCE AND TECHNOLOGY

Beijing, China

Aug 2016 - June 2020

- Minor in Finance and Entrepreneurship

Publications

CONFERENCE

Jinlin Lai, Charles Margossian, Daniel Sheldon. (2026). Corrected Integrated Laplace Approximation for Bayesian Inference in Latent Gaussian Models. In *Proceedings of the 40th Conference on Neural Information Processing Systems (NeurIPS, Spotlight, to appear)*. [preprint]

Max Hamilton, **Jinlin Lai**, Daniel Sheldon, Subhansu Maji. (2026). Scalable Model-Assisted Multi-Target Estimation in Large Image Collections. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*. [link]

Jinlin Lai, Antonio Ricardo Linero, Yuling Yao. (2026). Predictive Variational Inference: Learn the Predictively Optimal Posterior Distribution. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, Seoul, South Korea. PMLR 306, 2026. [link]

Max Hamilton*, **Jinlin Lai***, Wenlong Zhao, Subhansu Maji†, Daniel Sheldon†. (2025). Active Measurement: Efficient Estimation at Scale. In *Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS)*. [link]

Jinlin Lai, Justin Domke, Daniel Sheldon. (2024). Hamiltonian Monte Carlo Inference of Marginalized Linear Mixed-Effects Models. In *Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS)*, Vancouver, Canada. [link]

Jinlin Lai, Anustup Choudhury, Guan-Ming Su. (2024). Outdoor Scene Relighting with Diffusion Models. In *Proceedings of the 27th International Conference on Pattern Recognition (ICPR)*, Kolkata, India. [link]

Jinlin Lai, Javier Burroni, Hui Guan, Daniel Sheldon. (2023). Automatically Marginalized MCMC in Probabilistic Programming. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Honolulu, Hawaii, USA. PMLR 202, 2023. [link]

Jinlin Lai, Justin Domke, Daniel Sheldon. (2022). Variational Marginal Particle Filters. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS) 2022*, Valencia, Spain. PMLR: Volume 151. [link]

Haowen Xu, Wenxiao Chen, **Jinlin Lai**, Zhihan Li, Youjian Zhao, Dan Pei. 2020. Shallow VAEs with RealNVP Prior can Perform as Well as Deep Hierarchical VAEs. ICONIP.

WORKSHOP

Aditya Parashar*, Aditya Vikram Singh*, Avinash Amballa*, **Jinlin Lai**, and Benjamin Rozenoyer. (2024). "Quasi-random Multi-Sample Inference for Large Language Models." In *Frontiers in Probabilistic Inference: Learning meets Sampling*. [link]

Jinlin Lai, Daniel Sheldon. 2022. Automatic Inference with Pseudo-Marginal Hamiltonian Monte Carlo. ICML workshop Beyond Bayes: Paths Towards Universal Reasoning Systems.

Jinlin Lai, Lixin Zou, Jiaying Song. 2020. Optimal Mixture Weights for Off-Policy Evaluation with Multiple Behavior Policies. Offline Reinforcement Learning Workshop at Neural Information Processing Systems.

Services

- Served as a reviewer for
- ICML 2022, 2025, 2026
 - AISTATS 2023, 2024, 2025, 2026
 - AABI (ProbML) 2023, 2024, 2026
 - NeurIPS 2024, 2025, 2026
 - ICLR 2025
 - ACM SIGPLAN SLE 2025
 - TMLR

Teaching Experience

Fall 2025	Algorithms for Data Science , Teaching Assistant	University of Massachusetts Amherst
Spring 2025	Introduction to Computation , Teaching Assistant	University of Massachusetts Amherst
Fall 2024	Advanced Algorithms , Teaching Assistant	University of Massachusetts Amherst
Spring 2024	Probabilistic Graphical Models , Teaching Assistant	University of Massachusetts Amherst
Spring 2023	Probabilistic Graphical Models , Teaching Assistant	University of Massachusetts Amherst
Spring 2022	Probabilistic Graphical Models , Teaching Assistant	University of Massachusetts Amherst
Summer 2019	Algorithms for High School Olympics , Lecturer	Nanchang, Jiangxi Province, China
Summer 2018	Algorithms for High School Olympics , Lecturer	Ganzhou, Jiangxi Province, China
Summer 2017	Algorithms for High School Olympics , Lecturer	Ganzhou, Jiangxi Province, China

Talks

Jinlin Lai. 2023. Automatically Marginalized MCMC in Probabilistic Programming. Contributed talk in *the 5th Symposium on Advances in Approximate Bayesian Inference*.